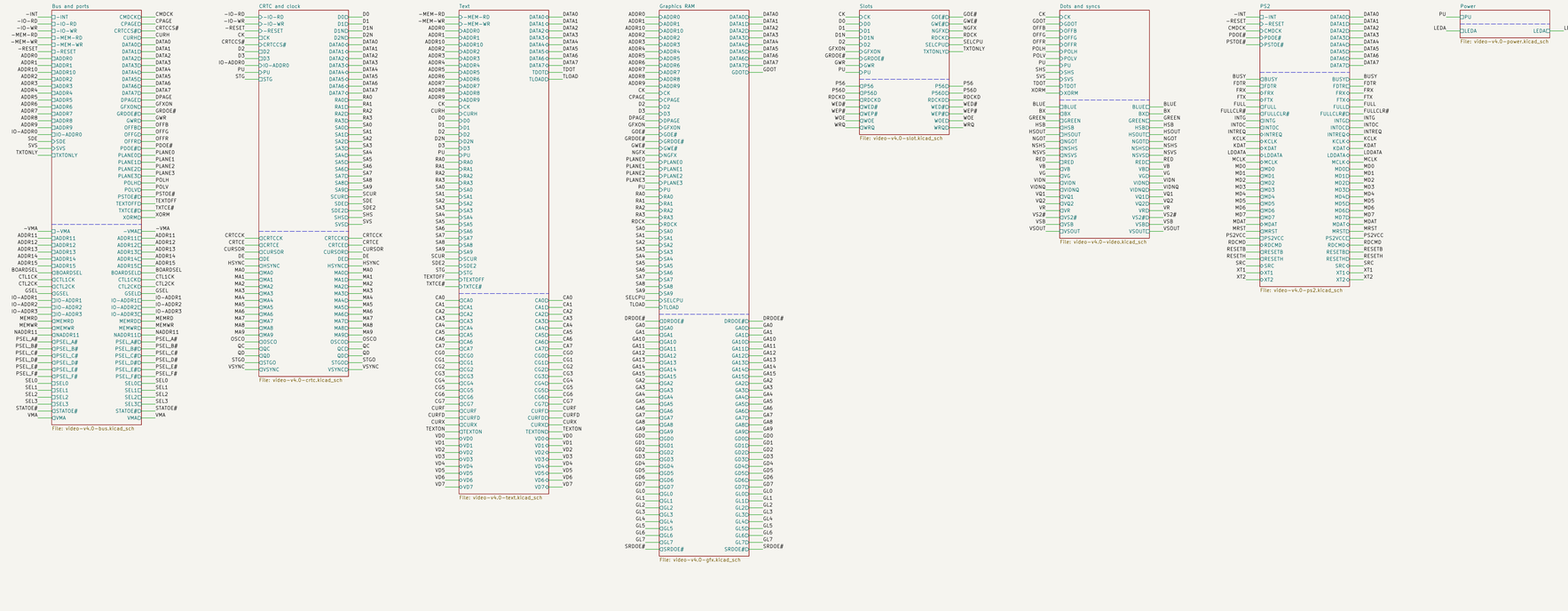
The video card V4.0, one sheet per functional block:
  sheet 2 Bus and ports BUS INTERFACE: THE BLOCK SELECT, THE SIX PORTS' STROBES, THE TWO CONTROL REGISTERS, THE STATUS
  sheet 3 CRTC and clock THE 6845, THE DOT CLOCK, THE DOT COUNTER, THE 6845'S CLOCK, THE STAGE REGISTERS
  sheet 4 Text   THE TEXT RAM, THE CODE REGISTER, THE CHARACTER GENERATOR, THE TEXT SHIFTER, THE CURSOR FLAG
  sheet 5 Graphics RAM THE GRAPHICS RAM: ADDRESS MULTIPLEXERS, WRITE BUFFER. THE TWO READ PATHS, BYTE LATCH. SHIFTER
  sheet 6 Slots   THE SLOT LOGIC: SELCPU, THE WRITE FLIP-FLOP, THE ~WE PULSE, THE RAM'S ~OE. THE READ LATCH'S CLOCK
  sheet 7 Dots and syncs THE TWO LAYERS MIXED, THE DOT RETIMED. THE THREE GUNS, THE SYNCS. THE VGA SOCKET
  sheet 8 PS2   PS/2: ATMEGA328P. LATCHES ON PD / PE. STATUS. FTDI. ~INT. RESET
  sheet 9 Power   POWER UNITS, DECOUPLING, POWER LED, THE F PARTS' PULL-UP

Every net of a sheet is a pin of its sheet symbol here, and the local label at each pin names the net
(/NAME, the names on the board). Above the dashed line: the nets the sheet shares with other sheets (left:
it reads them; right: it drives them). Below it: the nets drawn on that sheet only, a pin on each side
(KiCad's ERC counts a root label as connected only when another label of its name is on the root sheet).
VCC and GND are power symbols on the sheets.



Design note and theory: docs/cards/video-v4.md
Circuit source: v40_netlist.py: these sheets are generated by gen_v40.py
6845 at 16 dots a count on ports PA / PB, 640 x 480 VGA, text RAM and graphics window TA the $D0
YACC1

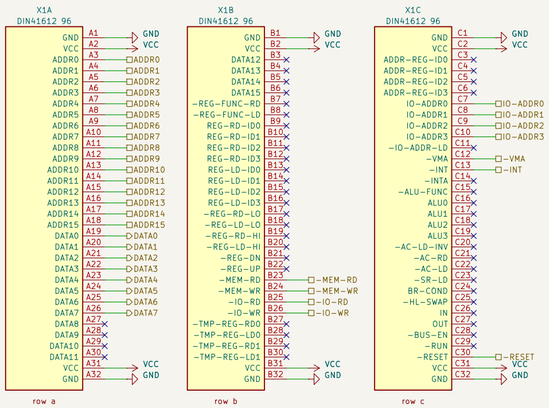
Sheet: /
File: video-v4.0.kicad_sch

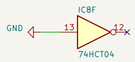
Title: YACC1 video card V4.0

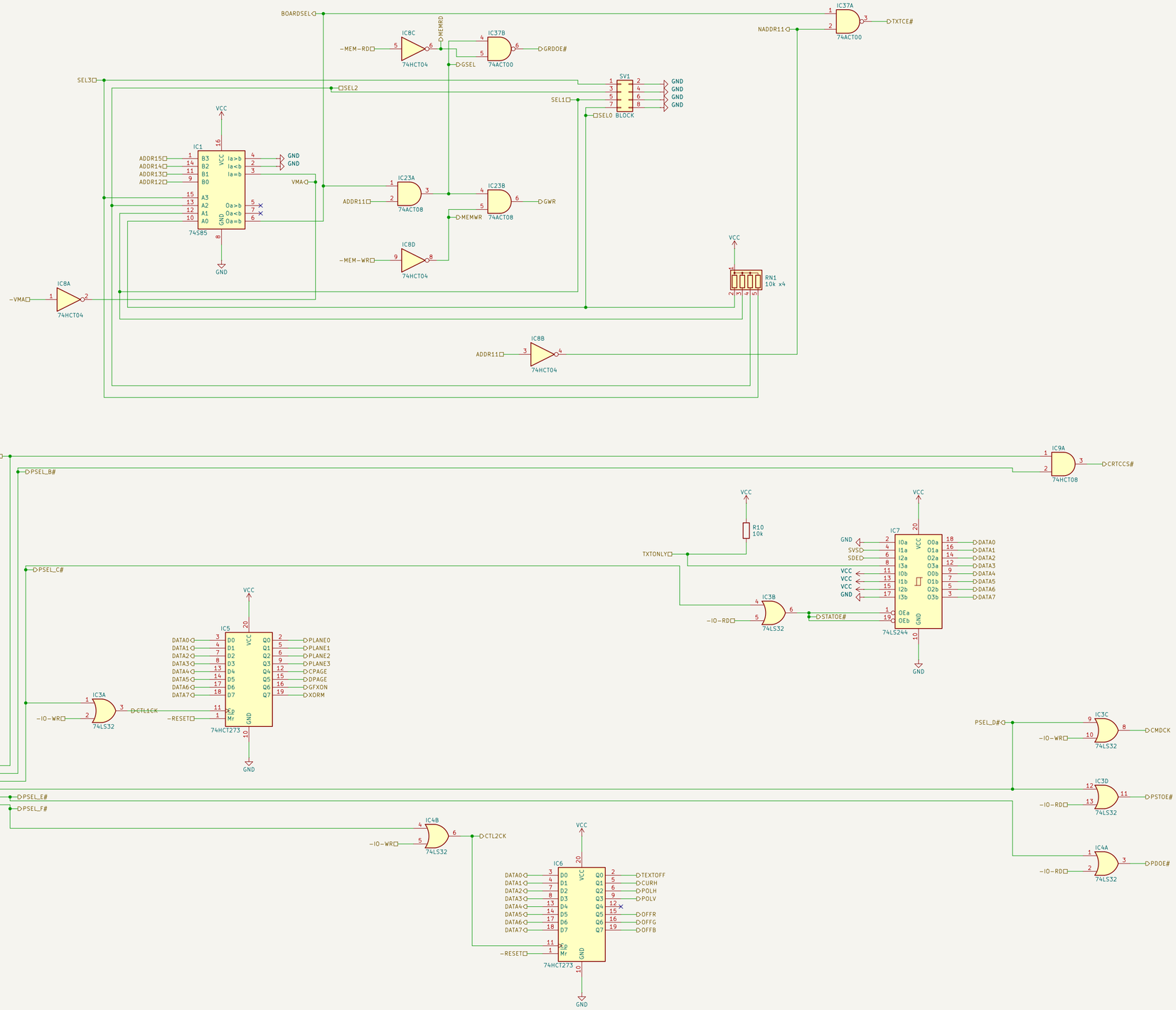
Size: A1	Date: 2026-11-13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 1/9

BUS INTERFACE: THE BLOCK SELECT, THE SIX PORTS' STROBES, THE TWO CONTROL REGISTERS, THE STATUS

X1: the YACC1 V3.2 bus (DIN 41612). IC1 compares ADDR15–12 with SV1 (5–6 fitted: the $D000 block). The block's two halves: TXTCE (IC37/A), the text RAM at $D000–$D7FF; GSEL (IC23/A), the graphics window at $D800–$DFFF, with its write strobe GWR and its read GRDOE. IC2 decodes IO-ADDR0–3: ports PA–PF = 1010–1111; IC3 / IC4 gate each select with a strobe. IC5 = control 1 (OUT PC), IC6 = control 2 (OUT PF), IC7 = the status (IN PC); the ID 0111.







Design note and theory: docs/cards/video-v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 2 of the video card V4.0 schematic (root: video-v4.0.kicad_sch)

YACC1

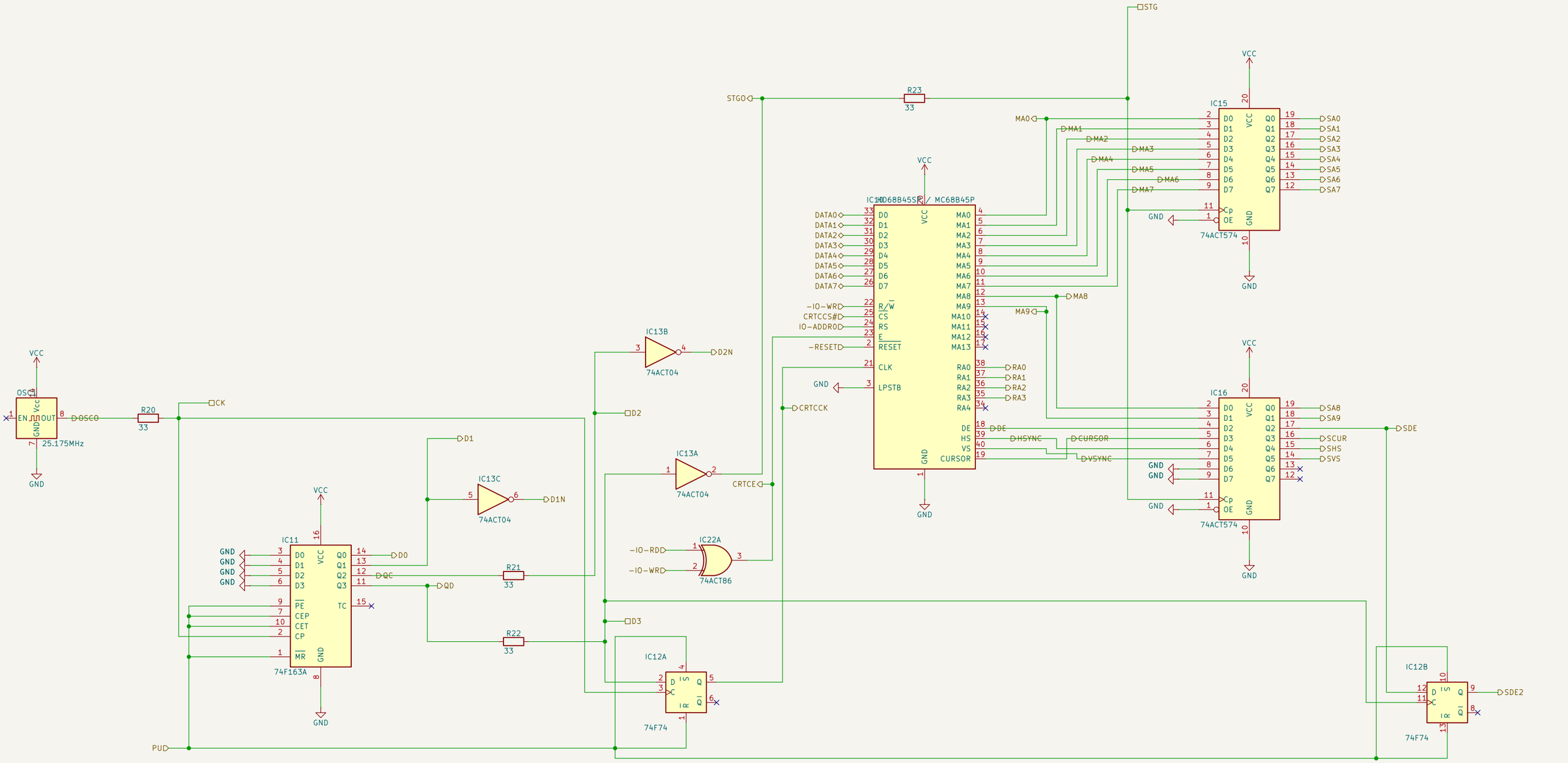
Sheet: /Bus and ports/
File: video-v4.0-bus.kicad_sch

Title: YACC1 video card V4.0: Bus and ports

Size: A2	Date: 2026-11-13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 2/9

THE 6845, THE DOT CLOCK, THE DOT COUNTER, THE 6845'S CLOCK, THE STAGE REGISTERS

IC11 counts the 25.175 MHz dot clock: D0–D3, the 16 dots of a count. The 6845's clock is D3 a dot late (IC12/A): it steps as dot 1 begins. IC15 / IC16, the stage, take MA0–9, DE, CURSOR, HSYNC and VSYNC as dot 0 begins (STG = NOT D3), 15 dots later: SA0–9, SDE, SCUR, SHS, SVS, steady for a count. IC12/B is SDE2, SDE as dot 8 begins.
The 6845's bus side: –CS = PA or PB (IC9/A, sheet 2), E = –IO–RD XOR –IO–WR (IC22/A), RS = IO–ADDR0.
R20–R23 (33 ohm): series termination of CK, D2, D3 and STG at their sources.



Design note and theory: docs/cards/video–v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 3 of the video card V4.0 schematic (root: video–v4.0.kicad_sch)
YACC1

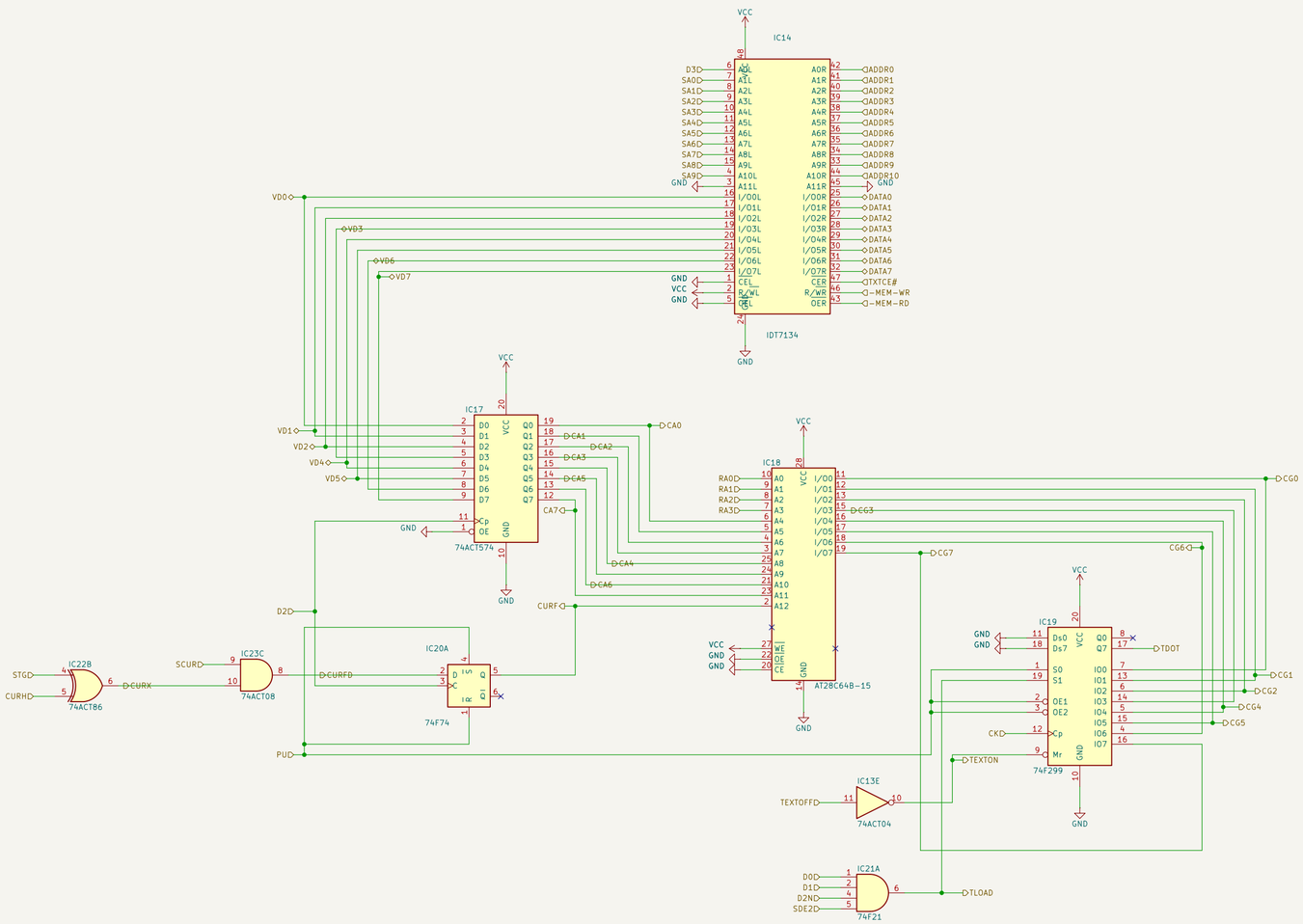
Sheet: /CRTC and clock/
File: video–v4.0–crtc.kicad_sch

Title: YACC1 video card V4.0: CRTC and clock

Size: A3	Date: 2026–11–13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 3/9

THE TEXT RAM, THE CODE REGISTER, THE CHARACTER GENERATOR, THE TEXT SHIFTER, THE CURSOR FLAG

The text RAM's display port reads SA9–0 then D3 (the half: 0 the left character, 1 the right). IC17 takes a code as dots 4 and 12 begin (D2), and IC20/A with it the cursor flag: SCUR AND (the half = CURH). The character generator IC18 is addressed by RA0–3, the code and the flag (A12: the cursor's forms); IC19 loads its byte 8 dots after the code was taken (TLOAD, IC21/A: dots 3 and 11 with SDE2) and shifts it out bit 7 first as TDOT.



Design note and theory: docs/cards/video–v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 4 of the video card V4.0 schematic (root: video–v4.0.kicad_sch)

YACC1

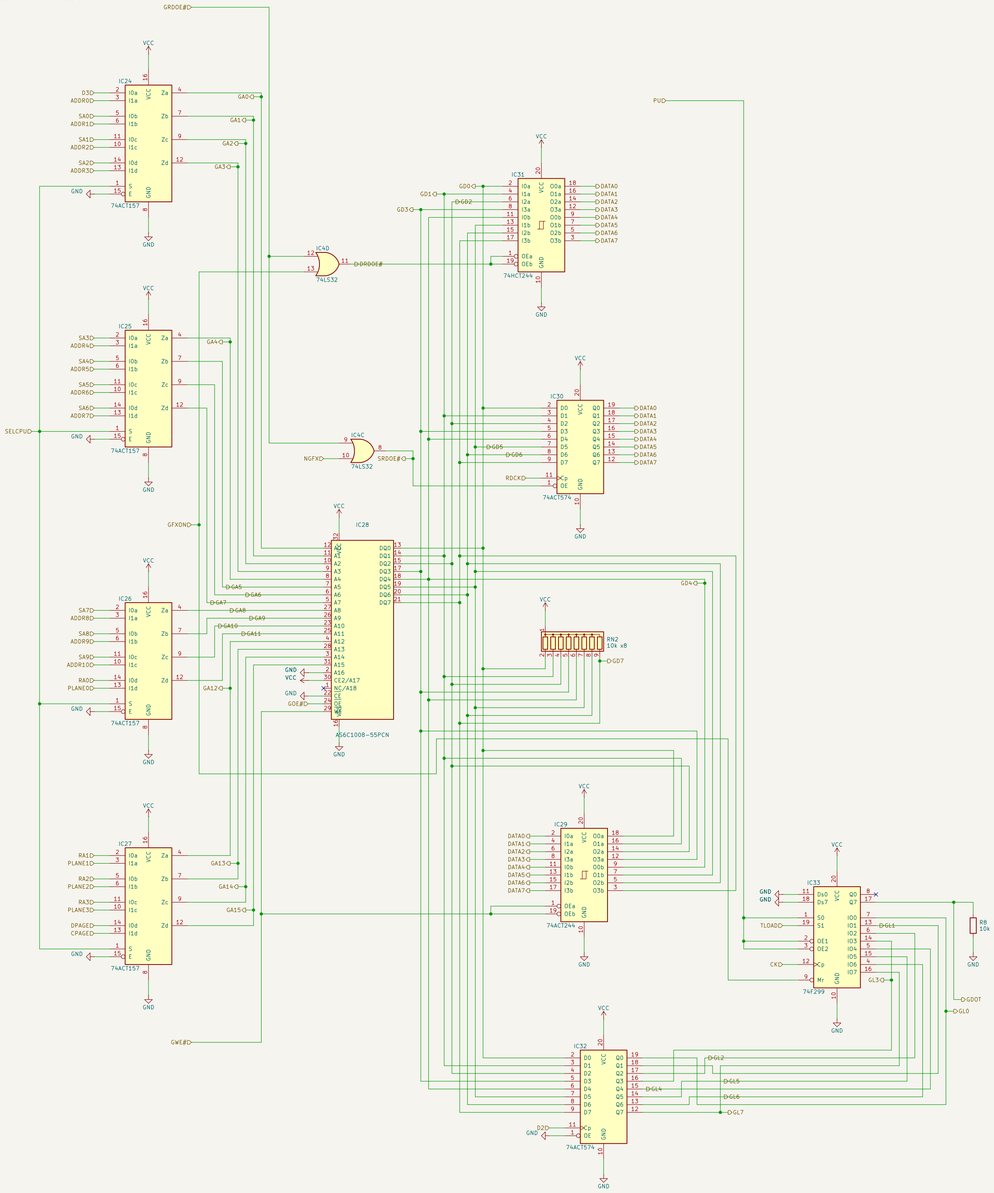
Sheet: /Text/
File: video–v4.0–text.kicad_sch

Title: YACC1 video card V4.0: Text

Size: A2	Date: 2026–11–13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 4/9

THE GRAPHICS RAM: ADDRESS MULTIPLEXERS, WRITE BUFFER, THE TWO READ PATHS, BYTE LATCH, SHIFTER

IC24–IC27 give the RAM the display's address (DPAGE, RA3–0, SA9–0, D3) or the CPU's (CPAGE, PLANE3–0, ADDR10–0): SELCPU. IC32 takes the display's byte on D2's rise and IC33 loads it with the text's (TLOAD). IC29 puts the bus on the RAM's data lines while ~WE is low. A window read: in slot mode from IC30, which takes the RAM's byte as each CPU slot ends (RDCK); in direct mode through IC31. These parts are the text-only build's left-off ones (with IC34, sheet 6).



Design note and theory: docs/cards/video-v4.md
Circuit source: v40_netlist.py: this sheet is generated by gen_v40.py
Sheet 5 of the video card V4.0 schematic (root: video-v4.0.kicad_sch)

YACC1

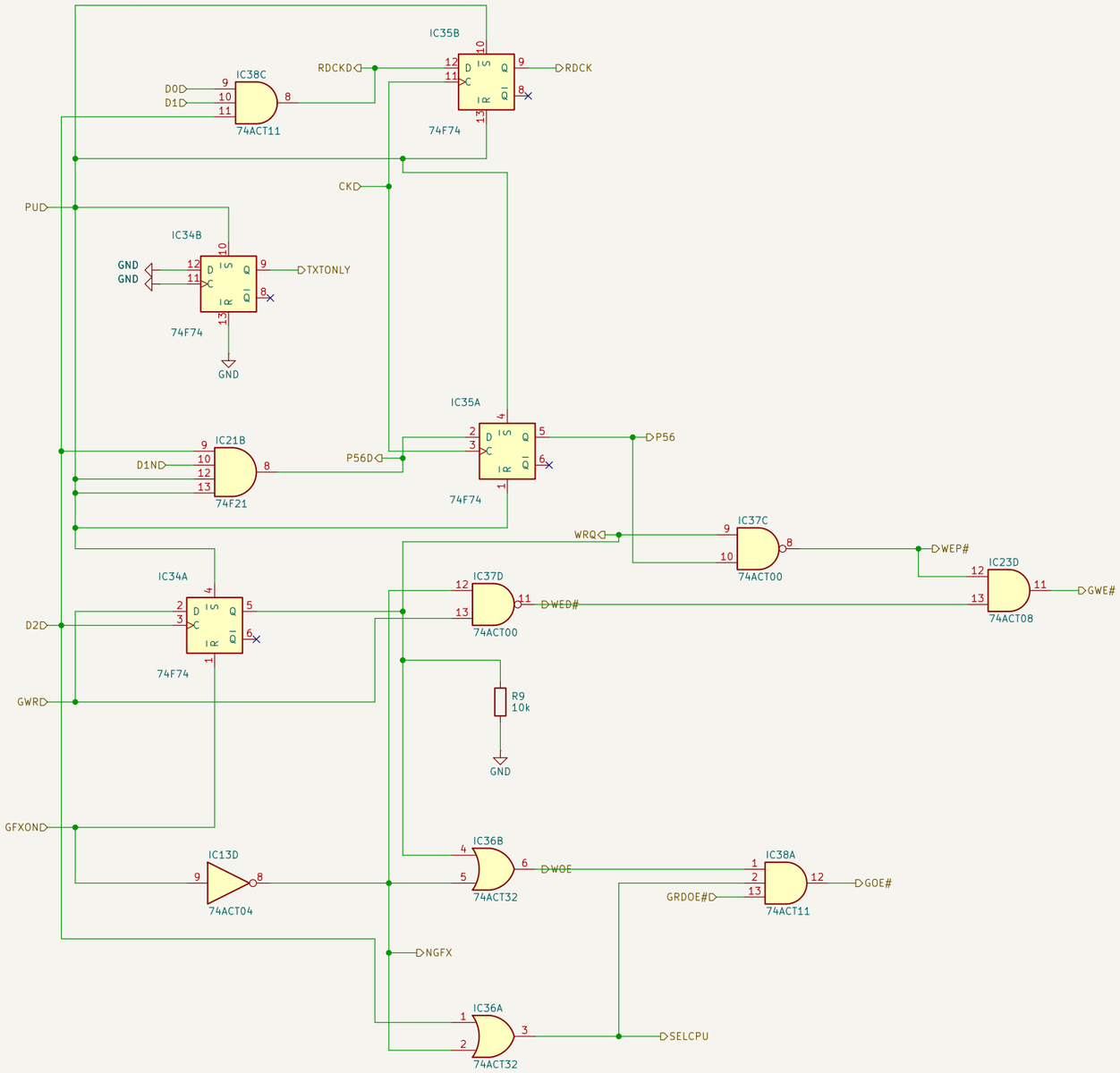
Sheet: /Graphics RAM/
File: video-v4.0-gfx.kicad_sch

Title: YACC1 video card V4.0: Graphics RAM

Size: A1	Date: 2026-11-13	Rev: V4.0
KiCad E.D.A. 10.0.6		Id: 5/9

THE SLOT LOGIC: SELCPU, THE WRITE FLIP-FLOP, THE -WE PULSE, THE RAM'S -OE, THE READ LATCH'S CLOCK

SELCPU = D2 OR NOT GFX-ON: with GFX-ON the display has the RAM in dots 0-3 and 8-11, the CPU in 4-7 and 12-15; without it the CPU has it for good (direct mode). WRQ (IC34/A) takes the window's write strobe as a CPU slot begins; P56 (IC35/A) is high in dots 5-6 and 13-14. -WE = NAND(WRQ, P56) AND NAND(NOT GFX-ON, GWR): a whole pulse in slot mode, the strobe itself in direct mode. GOE = (WRQ OR NOT GFX-ON) AND SELCPU AND GRDOE. RDCK (IC35/B) is high in dots 8 and 0. IC34/B holds TXTONLY low: the graphics parts are fitted.



Design note and theory: docs/cards/video-v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 6 of the video card V4.0 schematic (root: video-v4.0.kicad_sch)
YACC1

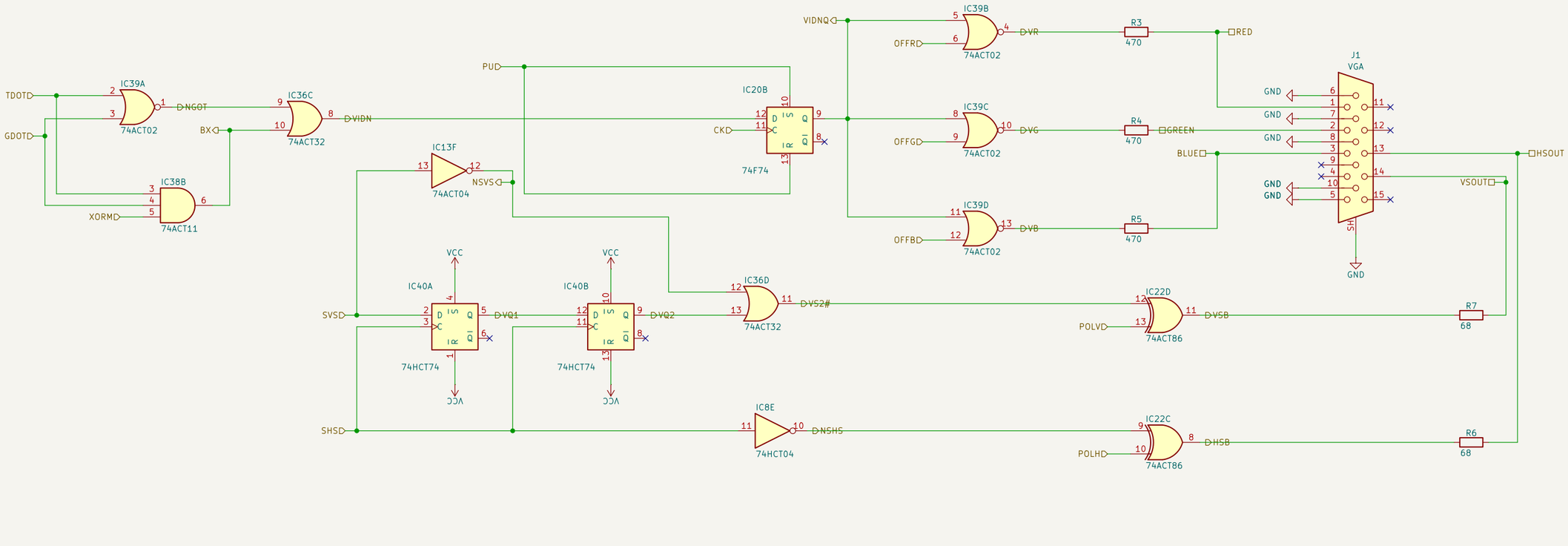
Sheet: /Slots/
File: video-v4.0-slot.kicad_sch

Title: YACC1 video card V4.0: Slots

Size: A3	Date: 2026-11-13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 6/9

THE TWO LAYERS MIXED, THE DOT RETIMED, THE THREE GUNS, THE SYNCS, THE VGA SOCKET

VIDN = NOR(text, graphics) OR (text AND graphics AND XOR); IC20/B puts it on the dot clock; each gun is NOR(VIDN, its off bit) through 470 ohm into the monitor's 75. HSYNC: the stage's, inverted, XOR POLH. VSYNC: IC40 takes SVS at each HSYNC and again at the next; the pulse to the monitor lasts from SVS's rise to the second HSYNC after it (VS2), whatever the 6845's own length; XOR POLV. Both negative after −RESET.



Design note and theory: docs/cards/video−v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 7 of the video card V4.0 schematic (root: video−v4.0.kicad_sch)
YACC1

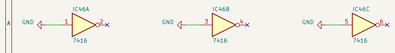
Sheet: /Dots and syncs/
File: video−v4.0−video.kicad_sch

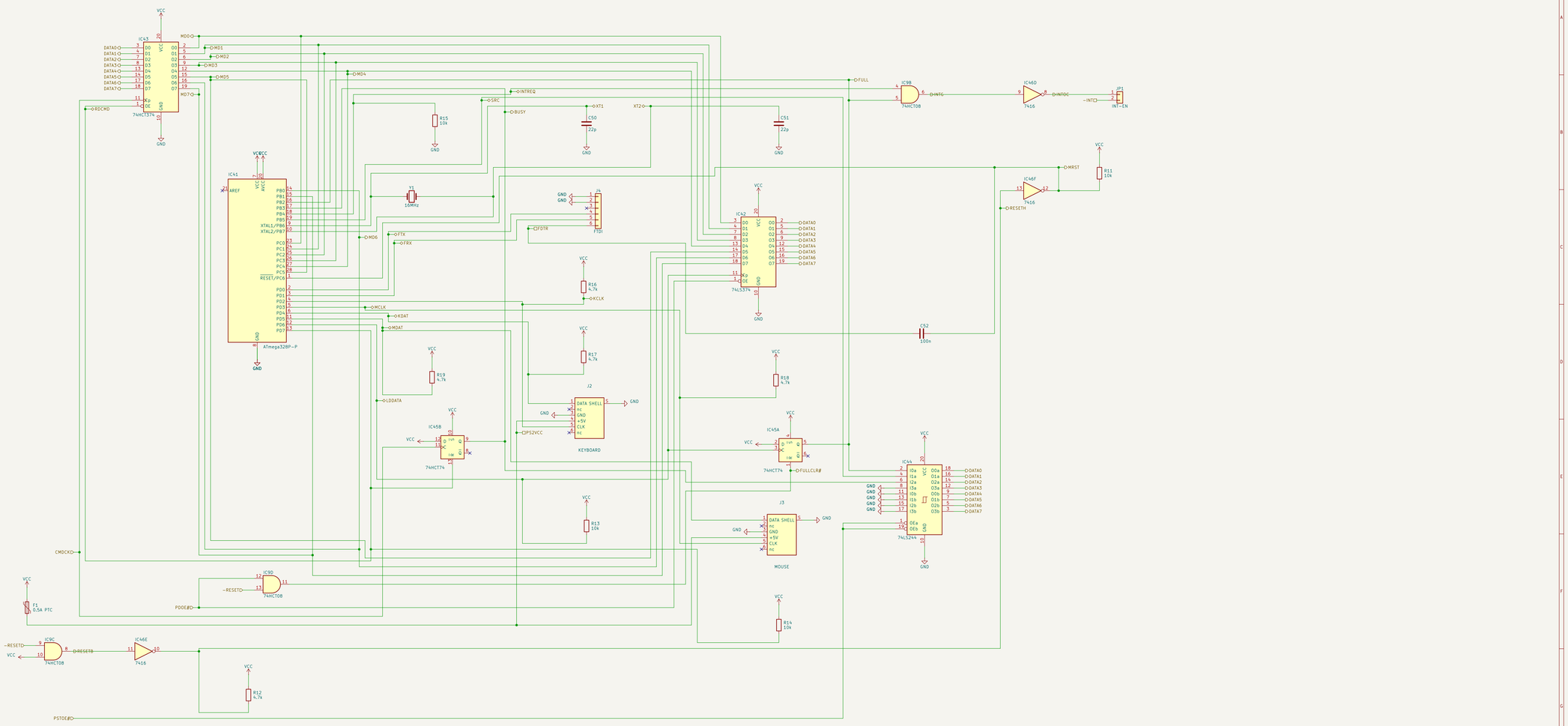
Title: YACC1 video card V4.0: Dots and syncs

Size: A3	Date: 2026−11−13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 7/9

PS/2: ATMEGA328P, LATCHES ON PD / PE, STATUS, FTDI, –INT, RESET

IC41 runs embedded/video-ps2. OUT PD loads a command into IC43 (BUSY set). IN PE reads a byte from IC42 (FULL cleared). IN PD the status IC44: FULL, SRC, BUSY, –INT = INTREQ AND FULL (IC9/8) through IC46/8 and JP1.
The bus –RESET reaches MRST through IC9/C, IC46/E and IC46/F (open collector, R12). It only ever pulls MRST low.
The V2.0 card's section, its nets unchanged.





Design note and theory: docs/cards/video-v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 8 of the video card V4.0 schematic (root: video-v4.0.kicad_sch)

YACC1

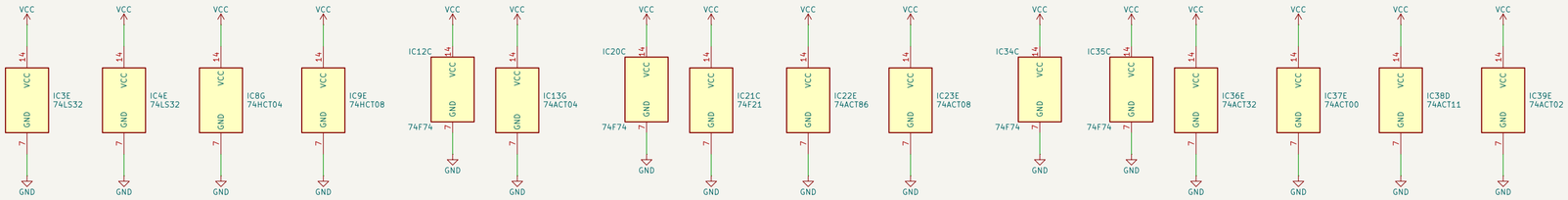
Sheet: /PS2/
File: video-v4.0-ps2.kicad_sch

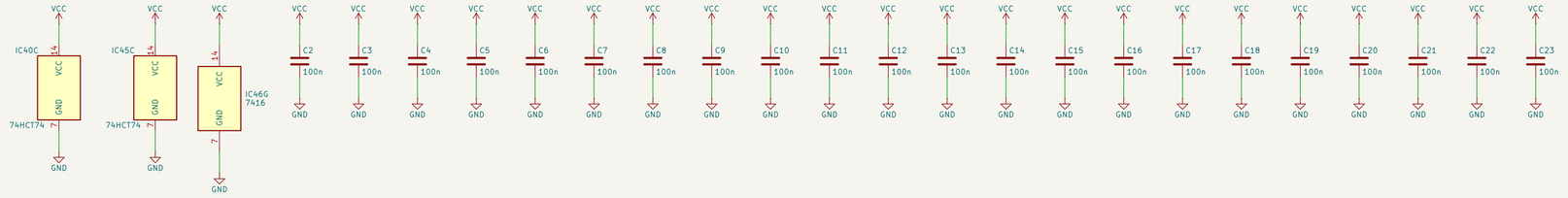
Title: YACC1 video card V4.0: PS2

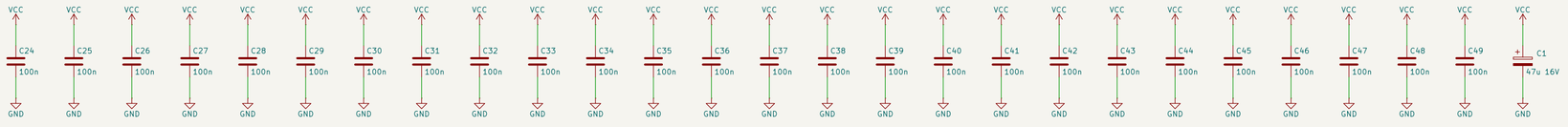
Size: A1	Date: 2026-11-13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 8/9

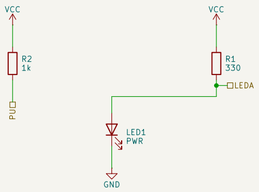
POWER UNITS, DECOUPLING, POWER LED, THE F PARTS' PULL-UP

One 100 nF per IC beside its VCC pin (C48 at the oscillator, C49 at the ATmega's AVCC), C1 47 uF bulk. The bus supplies VCC and GND (X1, six pins each): the PWR_FLAGs. PU (R2, 1k) holds the 74F parts' unused inputs high. Spare: one inverter of IC8, one OR of IC36 (inputs tied), three of IC46, control 2's bit 4.











PWR_FLAGs: the bus supplies VCC and GND (X1, six pins each).

Design note and theory: docs/cards/video-v4.md
Circuit source: v40_netlist.py; this sheet is generated by gen_v40.py
Sheet 9 of the video card V4.0 schematic (root: video-v4.0.kicad_sch)
YACC1

Sheet: /Power/
File: video-v4.0-power.kicad_sch

Title: YACC1 video card V4.0: Power

Size: A2	Date: 2026-11-13	**Rev: V4.0**
KiCad E.D.A. 10.0.6		Id: 9/9